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Sorting out the essence of owner–contractor collaboration in capital project delivery

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Abstract

Despite the relatively widespread recognition of relational-based contracting in engineering and construction projects, literature indicates a range of paradoxical issues in practice. This study attempts to reconstruct project practitioner's perspectives regarding the essence of collaborative relationships. Applying Q-methodology, subjective opinions and reflections of 30 project practitioners from 19 owner and engineering-construction firms were systematically analyzed. The result suggests four distinct perspectives towards effective working relationships, namely a) shared team responsibility, b) execution focused team, c) joint capability and structure; and d) senior leadership pair. Across perspectives, all practitioners shared a belief that an effective owner–contractor relationship should be based on affective trust, shared vision, and mutual attitudes such as open and honest communication, solution seeking instead of blaming, and senior management leadership. In contrast to prior research, long-term orientation and contractual functions were perceived to play a relatively limited role in improving owner–contractor relationships.
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1. Introduction

The nature of working relationships between owner and contractor in engineering and construction projects is considered to have a major effect on project performance (Drexler and Larson, 2000; Larson, 1995; Meng, 2011). If both parties can align their interests and develop a collaborative working relationship, potential conflicts can be dealt with before becoming claims, knowledge is freely exchanged, problems can be solved, and in turn parties can integrate their specific capabilities to complete the project successfully. On the other hand, adversarial or deteriorated relationships between project parties might lead to poor project performance (Black et al., 2000; Humphreys et al., 2003; Meng, 2011). Under adversarial relationships, owners are likely to challenge requests for approval, force compliance by withholding funds, and overly control the contractors' works. Meanwhile, contractors might exploit potential claims by aggressively negotiating change orders and withholding vital information. Often, small issues easily escalate into major disputes causing costly delays and ending in formal litigation.

In the light of improving project performance, focus on developing and practicing better ways of working between owners and contractors has intensified. This resulted in prescriptive models of relational contract such as project partnering and project alliancing (see Anvuur and Kumaraswamy, 2007; Bygballe et al., 2010; Chan et al., 2003; Kumaraswamy et al., 2008). In essence, such prescriptions aim at developing a collaborative relationship characterized by quality such as aligned goals and interests, open and honest communication, mutual commitment and trust, long-term orientation, and joint problem solving (Anvuur and Kumaraswamy, 2007; Bresnen and Marshall, 2000; Rahman and Kumaraswamy, 2004a).

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In one stream of research, project partnering or alliancing was reported to be practiced successfully (Barlow, 2000; Bayliss et al., 2004; Davis and Walker, 2008; Larson, 1995). Larson's (1995) examination of 280 construction projects suggests that partnered projects were more successful in controlling costs and resulted in better performance and customer satisfaction than projects managed under more adversarial conditions. Based on the BP Andrew Alliance case, Barlow (2000) claims that the partnering approach had not only delivered very significant performance gains, but also brought in technical and process innovation.

Another stream of research however also revealed the difficulties encountered in such practices. Bresnen and Marshall (2002) highlight that current practices might put too much emphasis on formal mechanisms (i.e. contracts, tools and techniques). Formalized practice often trivializes the important social dimensions of collaboration and the dynamics of relationships among different individuals within the permanent organizations and the project organization. Similarly, in Hong Kong construction projects, Chan et al. (2006; 2003) find the major obstacle of partnering success to be the failure of introducing desirable partnering attitude due to commercial pressure. Based on a case study of two projects in the UK, Alderman & Ivory (2007) highlight the 'partnering paradox' where the intended collaborative behavior is hardly realized mainly due to the existence of adversarial attitudes. A study of five projects in Norway and Canada (Aarseth et al., 2012) confirms the existence of 'practical difficulties' due to a lack of collaborative mind-set and insufficient initial effort to establish shared norms.

The authors aim to extend current literature on project-based collaborative relationships by empirically reconstructing the project practitioner's perception on essential ingredients to improve the owner–contractor relationships. The main question is: how would practitioners improve their working relationships to ensure successful project delivery? This paper is structured as follows: firstly, we discuss the concept of collaborative relationship — what we know about it and its essential elements. Then, we briefly discuss our approach, based on Q-methodology, to data collection and systematic analysis of the project practitioner's opinions. Next, we systematically present the revealed practitioners' perspectives on the essence of improving owner–contractor relationships. Finally, we discuss the practical and theoretical implications of the results.

2. Owner–contractor relationships in projects

The effects of the nature of owner–contractor relationships have been investigated in a series of project partnering studies (Chan et al., 2004; Drexler and Larson, 2000; Larson, 1995; Meng, 2011). There is a mixed result in project performance when some degree of partnering (*informal partnering*, *project partnering*, and *strategic partnering*) was used to measure the owner–contractor relationship efficacy. Almost two decades ago, Larson (1995) reports superior project performance achieved by both informal-partnered and fully formal-partnered projects. Recent findings by Meng (2011) however, show that partnered and even strategic partnered projects did not significantly differ from projects managed by traditional relationship. When the relationship quality indicators were used to measure the effect of owner–contractor relationship, more convincing results emerged. Based on 100 respondents, Meng (2011) concludes that schedule performance can be improved by joint working; cost performance can be improved through open and effective communication, clear and fair risk allocation, regular performance measurement, and no-blame culture; and quality defects can be reduced through effective problem solving mechanisms.

Building on Meng's findings, our focus is on investigating the essential ingredients of owner–contractor collaborative relationships, and not on the effect of different levels of relationships on the project performance. Extant literature suggests two different types of relationship in projects: *adversarial* and *collaborative* (Cheung et al., 2009; Humphreys et al., 2003; Smyth and Pryke, 2008b). It is clear that an adversarial relationship is undesirable and assumed to lead to short-term, opportunistic behavior and confrontational interactions between owner and contractor. A collaborative relationship, on the other hand is desirable and generally characterized by commitment, cooperation, and connectedness of owner and contactor striving for a common goal. However, collaborative relationships may also take a range of approaches such as *strategic alliances*, *project partnering*, *project alliancing*, and *supply chain management* (BSI, 2010; Bygballe et al., 2010; Xue et al., 2010); *relational contracting* (Gil, 2009; Rahman and Kumaraswamy, 2005, 2008); and *integrated teamworking* (Baiden and Price, 2011; Baiden et al., 2006; Bosch-Rekveldt et al., 2011; Kumaraswamy and Rahman, 2006; Thomas and Thomas, 2005). We therefore define owner–contractor collaborative relationship in a project as the behavioral interaction between owner and contractor working together for the purpose of achieving specific project and business objectives by effective utilization of each party's specific resources and capabilities based on shared values and norms.

Based on a review of related literature, we found an extensive list of elements or components of collaborative relationships. We further identified six key categories emerging from literature, namely: (i) teamworking, (ii) relational attitudes, (iii) capability, (iv) team integration, (v) joint working, and (vi) contract. These six categories are chosen arbitrarily to reflect different conceptual angles on what constitutes collaborative relationships. These categories are not meant to classify different types of relationships (as the terms such as partnering, alliance, or strategic alliance do) but are considered as high order factors of the elements of collaborative relationship. The six categories will be further detailed in the next six paragraphs.

2.1. Teamworking

The role of teams to the success of organizations is well documented in the management literature (Ilgen et al., 2005; Mathieu et al., 2008; Stewart, 2010). In project context, teamworking can be defined as the extent to which members in

a team work together on the basis of synergies in their interactions (Baiden and Price, 2011; Hoegl and Gemuenden, 2001; Tannenbaum et al., 1992). The efficacy of teamworking to collaborative relationship is well recognized in the project management research (Baiden and Price, 2011; Chan et al., 2004; Cheung et al., 2009; Hoegl and Gemuenden, 2001; Jefferies et al., 1999; Kumaraswamy and Rahman, 2006). Cheung et al. (2009) found that good teamwork spirits is one of the important drivers for the contracting parties to adopt cooperative behavior. In a series of empirical studies, Hoegl and colleagues (Hoegl and Gemuenden, 2001; Hoegl and Parboteeah, 2006; Hoegl and Parboteeah, 2007; Hoegl and Weinkauf, 2005; Hoegl et al., 2004) show that 'teamwork quality' significantly contributes to the success of innovative projects. Building upon prior empirical studies we identified a number of common elements of teamworking: team identity or cohesion, shared vision, information/knowledge sharing, team member's affective trust, attitude towards diversity in problem solving, and reflection and self-assessment.

2.2. Relational attitudes

Different attitudes and mindsets exist within each party prior to any relationship between the two parties and are brought into the relationship. As the two parties interact, a set of relational norms, factors, or routines are co-developed specific to govern their relationship (Poppo and Zenger, 2002; Poppo et al., 2008; Rahman and Kumaraswamy, 2008; Spekman et al., 1998). In this study, we position relational attitudes as a high order concept reflecting the following relational norms and aspects between owner and contractor: inter-organizational trust alongside organizational cultural fit, open communication, long-term orientation, and top management commitment. Inter-organizational trust and open communication is evidenced to improve buyer–supplier relationships (Poppo and Zenger, 2002; Ring and van de Ven, 1994; Zaheer et al., 1998) and project performance (Pinto et al., 2009). While long term orientation (Ahola, 2009; Ganesan, 1994), organizational cultural fit and top management commitment (Black et al., 2000; Rahman and Kumaraswamy, 2008) also have been identified as critical success factors for collaborative relationship.

2.3. Capability

In projects, capability refers to both the owner's and contractor's project management capability (Turner, 2009), technical capability in a specific area (Rahman and Kumaraswamy, 2008), financial strength (Black et al., 2000), and perceived organizational reputation (Kadefors et al., 2007; Pesämaa et al., 2009). It is widely recognized that the contractor's capabilities are the basic requirement for delivering a successful project. Therefore, current project procurement procedures are designed to select the best capable contractor from the market pool (Morris and Pinto, 2007; Pesämaa et al., 2009; Pryke, 2009; Turner, 2003; Walker and Hampson, 2003). However, with increasing project complexities, owner (in-house) capabilities also become crucial to ensure project performance (Berends, 2007; Miller and Lessard, 2000; Rahman and Kumaraswamy, 2008).

2.4. Team integration

Team integration often refers to a collection of practices, methods and behaviors that promote a favorable environment where information and knowledge are exchanged freely among the parties (Baiden and Price, 2011; Bosch-Rekveldt et al., 2011; Dietrich et al., 2010; Merrow, 2011). Team integration has common features: (1) creation of a single integrated project team, (2) seamless operation without organizational boundary, (3) unrestricted cross-sharing of information, (4) equitable relation and respect for all, and (5) collective responsibility for all project outcomes (Baiden and Price, 2011). The investigation by Bosch-Rekveldt et al. (2011) on 5 projects suggests that integrated teams are more open to share information and knowledge and are therefore more effective in anticipating changes in the highly dynamic project environment.

2.5. Joint working

Instead of confronting each other, a collaborative relationship supposedly enables the parties to make joint efforts for managing the project tasks. Joint working is generally reflected by: (1) joint decision making (Chan et al., 2004); (2) joint problem solving and dispute handling (Baiden and Price, 2011; Black et al., 2000; Chan et al., 2004; Rahman and Kumaraswamy, 2008); (3) joint risk management (Bosch-Rekveldt, 2011; Rahman and Kumaraswamy, 2002; Rahman and Kumaraswamy, 2004a) and (4) joint effort for continuous improvement (Black et al., 2000; Chan et al., 2004; Meng, 2011; Rahman and Kumaraswamy, 2008).

2.6. Contract

Literature categorizes contracts and their key aspects such as remuneration scheme, incentives and risk sharing mechanism in a broad range of terms. The type and scope of a contract is typically chosen by the owner, depending on the nature of the works, availability of own resources, availability of contractors, project value drivers, and required competences. Generally, a contract specifies roles, responsibilities, remuneration scheme, payment terms and phases, incentive scheme, distribution of risk, and dispute resolution, and conflict settlement (Turner, 2003; Turner and Simister, 2001; von Branconi and Loch, 2004). So far, the views on the effectiveness of incentive based and partnering contracts on owner–contractor working relationships compared to lump-sum contracts are contradictory. On the one hand, partnering contracts are considered more collaborative than lump-sum based contracts (e.g.:Bayliss et al., 2004; Larson, 1995). On the other hand, it is suggested that partnering does not always eliminate the underlying adversarial attitude (Bresnen and Marshall, 2002; Ng et al., 2002). Merrow (2011, p.292) reinforces this view that "designers of alliances also thought that they would get the best features of lump-sum contract with none of the draw backs. […] what we actually end

up with [is] the worst features of lump-sum contracts combined with the worst features of reimbursable contracts".

It is important to note here that there are interrelationships between some conceptual categories. Extant literature suggests that teamworking reflects the effects of relational attitudes, team integration, and joint working. For example, Baiden and Price (2011) suggest that team integration can help improve teamwork effectiveness, however, they also recognize that team integration is not the only supporting factor to teamwork effectiveness. Their study shows that "different levels of integration can result in the same levels of teamwork effectiveness" (p.135). Practicing team integration is only one factor among others, whether or not teamworking can be achieved further depends on the inherent mechanisms of teamworking. We therefore consider the categories separately because each category represents a key area of a collaborative relationship. Moreover, because our focus is to gain insights from the practitioners' on how to improve the working relationship between owner and contractor in the execution phase, we opt for comprehensiveness to allow us to select elements of collaborative relationship – as complete as possible – meaningful to the practitioners.

3. Methods

Taking a view that "projects are about state of mind" of the project practitioners (Winch and Maytorena, 2011, p.360), this study focuses on the practitioners' subjectivity. By subjectivity we mean that a practitioner's perception stems from his/her "internal frame of reference" that was casted through the experience of carrying role in different situations (McKeown and Thomas, 1988). A person's frame of reference also includes a person's belief that guides his/her own behavior and decision making (Cross, 2005).

We apply a systematic analysis of the project practitioners' subjectivity through application of Q-methodology. Q-methodology is an inverted technique of factor analysis (R-method), invented by psychologist–physicist William Stephenson in the 1930s (Stephenson, 1953) as the basis for a scientific approach to human subjectivity. The method can be used in any research field where subjectivity is at issue, including attitude measurement (Stephenson, 1953, 1965). Fundamentally it is a discursive, constructivist approach that combines a strong qualitative dimension with the powerful quantitative tool (Stenner et al., 2008). Mainly due to the work of Brown (1980) and McKeown and Thomas (1988), who further developed the principles and procedures of Q-sorting, the Q-methodology has more recently found its way to a wide variety of research areas.

3.1. Q-methodology approach

In contrast to R-methodology (which is based on surveys and multivariate factor analysis), Q-methodology identifies respondents' responses within the context of the valuation of all statements presented (see Ten Klooster et al., 2008). Typically, respondents are presented with a sample of statements (the Q-set) derived from a population of issues (concourse) around the topic of interest. A number of *well-informed* respondents (the P set) are asked to rank-order the statements from their individual points of view. Rather than random sampling (and large sample sizes), Q-methodology relies on a small sample of purposively selected respondents. By sorting the statements respondents give subjective meaning to the Q-set and thus revealing their subjective viewpoint (McKeown and Thomas, 1988). The respondents' sorting data (the Q-sorts) are then factor analyzed to reveal similarities in viewpoint.

Q-factor analysis is an inversion of the conventional by-item factor analysis (Stephenson, 1965). Q-factor analysis correlates persons instead of test items (i.e. by-person factor analysis). If each individual had his own specific preferences (likes and dislikes or agree and disagree), their Q-sorts would not correlate. If, however, significant clusters of correlations exist, they can be factorized and described as common viewpoints, and individuals can be mapped to a particular factor. Q-methodology can thus be used to reveal and describe populations of viewpoints rather than populations of people as in conventional factor analysis (Brown, 1980; Cross, 2005; Ten Klooster et al., 2008; Van Exel and de Graaf, 2005). In a study based on Q-methodology, a small sample of respondents is sufficient and selected to represent the breadth of opinion in a target population (Brown, 1980, 1996).

This study was conducted in four steps. Firstly, by formulating Q-samples (samples of population of opinion), the actual research instrument and the basis of any Q methodological study. Secondly, by conducting Q-sorting sessions with samples of respondents. Thirdly, by factor analyzing all Q-sorts data to reveal the distinct perspectives. Ultimately, the results of factor analysis were interpreted to reconstruct and make sense of the revealed distinct practitioners' perspectives on ways to improve owner–contractor relationship.

3.2. Q-samples

The raw material of Q-samples originated from various opinions regarding factors or aspects for improving owner–contractor relationship. We collected various statements from 10 interviews with senior practitioners (project directors and senior project managers), popular articles from websites and blogs of project management and contracting professionals (e.g.: *The Project Management Network* and *Commitment Matters blog*), popular literature (e.g.: ACA, 1999; CII, 2011; KPMG, 2010; Merrow, 2011), and journal articles (e.g.: Ahola, 2009; Bosch-Rekveldt, 2011; Bresnen, 2007; Chan et al., 2012; Meng, 2011; Smyth and Pryke, 2008a).

The raw material was edited and then categorized. Composite statements were split so that each addressed a single issue but broad and clear enough to be interpreted in slightly different ways by different people; similar statements were grouped and combined. After 7 iterative discussions, we compiled 115 statements. All statements were arbitrarily selected in accordance to six categories of collaborative relationship ingredients as discussed in Section 2: (i) teamworking (TW); (ii) relational attitudes (RAT); (iii) capabilities (CAP); (iv) team integration (TIN); (v) joint working (JOW); and (vi) contractual aspects (CTR). Finally, within each category we made a broadly

representative selection leading to a Q-set of 55 statements (see Appendix 1). Each statement was randomly assigned a number and printed on a card. To ensure operability of the instrument, we conducted a pilot Q-sorting test with 3 project management scholars resulting in minor editorial adjustments to the final statements.

3.3. Q-sorting

As the nature of working relationship involves different roles of practitioners and firms, we identified a number of respondents based on their roles in projects (project director, project manager, project engineer) and their firms positions (owner and contractor). Potential respondents were then approached by email or telephone to be sure they were willing to participate and were exposed to the working relationship with either owner or contractor.

Project practitioners who met the selection criteria and agreed to participate were invited to 60–90 minute face-to-face meetings. Each participant was asked to perform Q-sorting. Respondents were asked to rank-order all 55 statements (printed on cards) in an ordinal scale from −5 (most disagree) via 0 (neutral) to +5 (most agree) and whereby the guiding question was: "in order to improve owner–contractor relationship it is important that …". They were asked to sort the statements in relation to each other. Respondents put the cards on the forced normal distribution that was printed on a sheet of paper as show in Fig. 1.

Immediately after finishing Q-sorting, the participants were interviewed. During this post-sorting interview, the participants were asked some personal background questions about their career, education, and working experience. They were also invited to describe their latest completed project characteristics like project definition, contract type, performance with regard to budget, schedule and quality, and critical incidents occurred. These questions were presumed relevant for supporting qualitative interpretation of the participants' Q-sorting.

3.4. Q-factor analysis

To reveal factors, that is, the distinct ways in which the statements were rank-ordered, Q-sorts data from 30 respondents were factor analyzed using *PQMethod version 2.33* (Schmolck, 2012). We chose *principal component analysis* and *Varimax rotation* methods to extract and rotate factors. For each resulting factor (i.e. each different perspective for improving owner–contractor relationship), an array of *Z-scores* (Z-SC) was computed based on the Q-sort scores of the respondents loading on that factor and using their correlation coefficient with the factor as weight. An *idealized Q-sort* was reconstructed for each factor in which all 55 statements were recoded to *idealized Q-sort value* (Q-SV) ranging from +5 to −5 according to their Z-scores ranks. Each factor was interpreted and described using the characterizing and distinguishing statements and the explanations of respondents loading on the factor. A statement is *characterizing* if its position is in the outer columns of the idealized Q-sort of the factor (see Fig. 1) and *distinguishing* if the position is statistically significantly different from its position in the idealized Q-sorts of all other factors. Some explanations of respondents describing a factor are quoted in the results section to understand their way of thinking and to support the description of that particular viewpoint.

4. Results

4.1. Characteristics of respondents

A total of 30 practitioners (see Appendix 2) from 19 different companies were included as Q-sorters, i.e. 15 owner practitioners represented 9 different owner companies and 15 contractor practitioners represented 10 different engineering contractor companies. The practitioners were balanced between owner and contractor sides based on their positions. On the owner side, we included 4 project directors, 10 project managers, and 1 project engineer as the respondents; and on the contractor side 3 project directors, 10 project managers, and 2 project engineers. The practitioners' years of experience in projects were relatively comparable. The average years of practitioners' experience was 22.9 years ranging from 3 to 40 years.

The diversity of practitioners' experience was based on their last projects contract type and location. Of the 30 practitioners, 19 were involved in their last projects based *EPC* (Engineering, Procurement, and Construction) *lump sum* contract while the other 11 practitioners were involved in different contract types ranging from *EPCm* (Engineering, Procurement, and Construction management) *unit rate* (3), *EPCm convertible lump sum* (2), *project alliance*/*partnering* (4), to *long-term alliance* (2). In terms of the location of their last project, 20 practitioners conducted their projects in Europe and the rests were in Asia (3), Australia (1), Russia (1), and South America (1). Note that, the practitioners' last projects' contract type and location were relatively unbalanced. Typically, this is not important for this study since Q-methodology focuses on representativeness of the population of opinions instead of population of people (Brown, 1980; McKeown and Thomas, 1988).

4.2. Q-factor analysis

To determine the number of meaningful factors, i.e. the number of meaningful and distinct practitioner perspectives on ways to improve owner–contractor relationships, we analyzed the results of different consecutive factor rotations on two- to seven-factor solutions. For each factor solution, the following rules were applied to accept the solution (Brown, 1980):

Each acceptable factor should be defined by at least two significant Q-sorts, whereby:

(i) A Q-sort x is loaded significantly at $p < .05$ on a factor y *if* its factor loading, $f_{xy} > 0.27$ (*calculated from* $1.96\sqrt{N}$; where $N = 55$, *number of statement*); and
(ii) Its highest square factor loading explains more than half of the common variance, $f_{xy}^2 > h_x^2/2$; where h_x^2 is the sum of the square factor loadings of the Q-sort x.

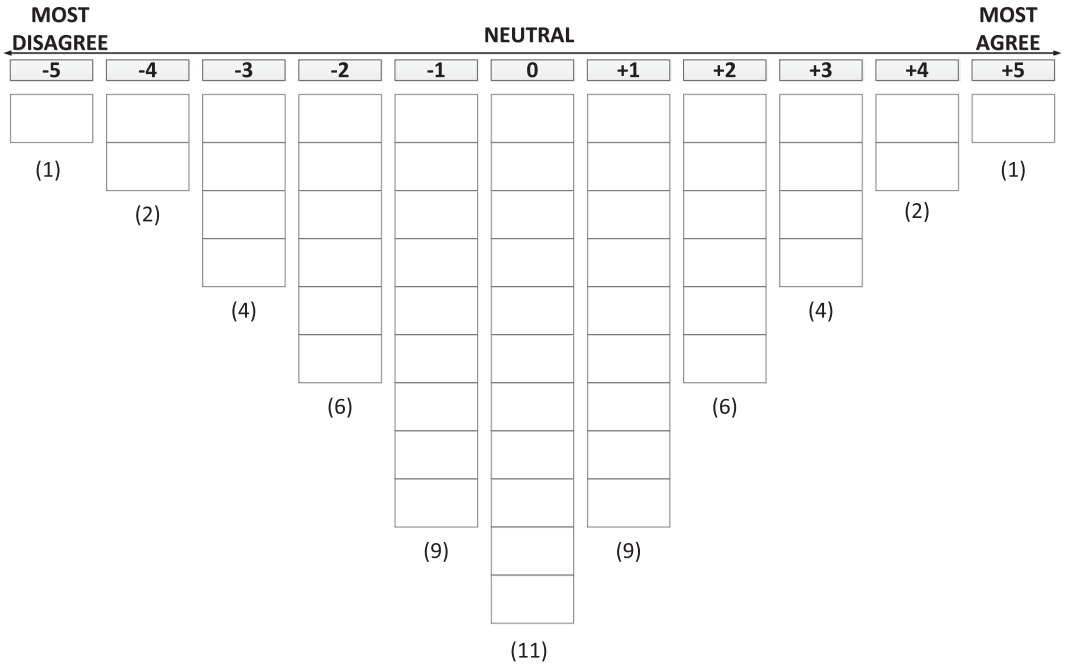


Fig. 1. Q-sort score board.

The result in Table 1 indicates that our data supports a maximum of 5 factors. Factor loadings of six- and seven-factor solutions resulted in only 5 acceptable factors (defined by at least two Q-sorts). Cumulatively explained variance (CEV) of four- and five-factor solutions is 55% and 60% respectively. Furthermore, the number of defining sorts for four- and five-factor solutions is respectively 24 and 23 sorts respectively. This means four- and five-factor solutions are relatively indifferent.

After interpreting the content characteristics of the factors in three- to five-factor solutions, we identified four stable distinct factors. The cumulative explained variance of 55% from four-factor solutions is also comparable to other Q-methodology studies (Ten Klooster et al., 2008; Van Exel, 2011). The factor loadings of four-factor solution are given in Appendix 3.

4.3. Practitioners' perspectives

To reveal the characteristics of practitioners' perspectives we used the results of factor analysis from four-factor solution (see Appendix 1). Firstly, quantitative outputs of Q-factor analysis, factor arrays were used to identify *characterizing* and *distinguishing* statements for each factor. Whenever we highlighted a statement, we included its identification number and corresponding Q-score value and placed it between parentheses. Furthermore, reflexive explanations from the corresponding defining respondents (R#, # denotes the respondent's identification code) were quoted to enhance our understanding on each perspective.

Table 1
Characteristics of consecutive factor solutions.

	2-Factor	3-Factor	4-Factor	5-Factor	6-Factor	7-Factor
CEV	44%	50%	55%	60%	65%	69%
Acceptable factors	2	3	4	5	5	5
Defining sorts	30	26	24	23	19	14

Source: compiled from consecutive Q-factor analysis using principal component analysis and *Varimax rotation*.

4.3.1. Perspective 1 (shared team responsibility)

The first perspective labeled as *shared team responsibility* accounts for 18% of the explained variance. Nine practitioners (2 owner project directors, 2 owner project managers, 3 contractor project directors, and 2 contractor project managers) have a common view with the array of statements for this perspective. Fig. 2 shows the factor scores of the characterizing statements for this perspective; the distinguishing statements of this perspective – the statements which corresponding factor scores distinct between this perspective and the others – are also highlighted.

These practitioners particularly emphasize that shared team responsibility is imperative to improve owner–contractor relationship, i.e. all the project team members should *share a common set of objectives* (2, +5) but stay *open minded on contentious issues and even conflicting opinions* (38, −5), and *embrace social interaction* (30, −4). Reflections from defining respondents encapsulate this core belief:

"Common objectives bind people and through this, people are prepared to accept their responsibilities and confirm what they want to achieve. In the end you will get the best out of

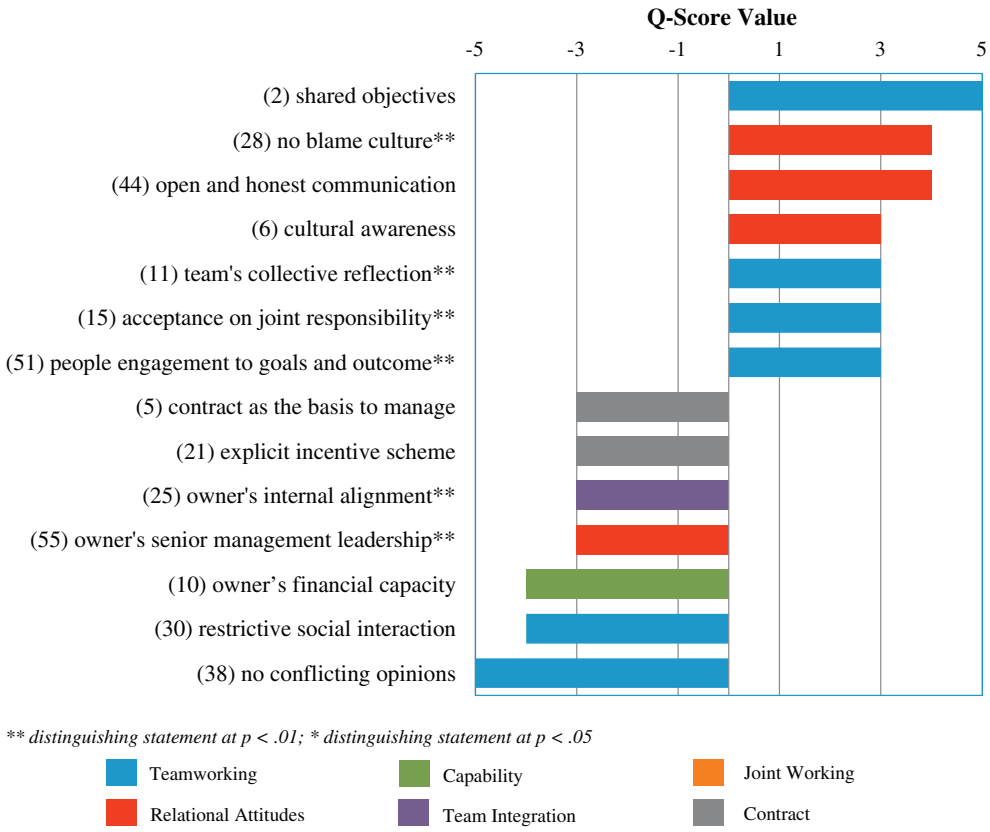


Fig. 2. Perspective 1 (shared team responsibility): defining and distinguishing statements.

them" (R2 and R13). "It will encourage people to take care of others to solve problems, taking it as individual and team responsibilities show your reciprocity in the relationship" (R23). "You need to have the same direction while allowing some differences to exist. Share, accept and be open about differences" (R25). "You need to embrace differences in opinions for a constructive discussion and mutuality" (R3, R23, and R28).

According to these practitioners, project teams with shared responsibility, should *regularly evaluate each other's roles and responsibilities* (11, +3), *accept joint* responsibilities for the team's achievement (15, +3), and *personally engage the team members to the project goals* (51, +3). To ensure such shared responsibility, both owner and contractor at firm level should embrace relational attitudes such as *no blaming each other but focus on finding solutions when problems occur* (28, +4), *open and honest* communication (44, +4), and acknowledge *cultural differences* (6, +3). The defining respondents argue this:

> "When you run into problems and any potential disputes, you need to solve it as one party and focus on what you can do best" (R10, R19, and R25). "Active communication without hidden agenda on both sides will be the basis for trust and improving the relationship" (R10, R25, and R26).
>
> "Project team is almost always multi-cultural (organizational and people), acknowledging this enables people to understand each other and in turn creates openness and honesty" (R23, R25, and R28).

Practitioners with this perspective also firmly believe that *owner's strong financial capacity* (10, −4) has nothing to do with the working relationship. "It is a given condition from selection process, you do not need this" (R2, R10, and R13). They particularly disagree that *owner's* (55, −3) as well as *contractor's senior management leadership* (8, −1) will improve the relationship. "In projects, leadership should be shown by the project manager; senior management does not need to be visible to people in the project team" (R23). They also reject the need to have *owner's internal alignment* (25, −3). "Owner's internal alignment is part of the business, only matters in the early stage and is not relevant during execution" (R10 and R13).

Interestingly, although not unique to this perspective is the use of a *contract as the basis to manage the project activities* (5, −3), and *explicit incentive schemes* (21, −3) might be hampering instead of improving the relationship. "In fact, in a real relationship you do not need a contract" (R10 and R19). "Incentive schemes should not be the main driver; instead making people work together should be based on respect" (R2 and R28).

4.3.2. Perspective 2 (execution focused team)

The second perspective labeled as *execution focused team* accounts for the 12% of explained variance and is defined by six practitioners (1 owner project director, 2 owner project managers, 2 contractor project managers, and 1 project engineer). As shown in Fig. 3 these practitioners underpin the project execution team with strong senior management leadership from both contractor and owner as the way to improve owner–contractor relationship. As in perspective 1, they firmly believe that everyone in the project team should *trust each other* (43, +5), *share a common set of objectives* (2, +4), *embrace social interaction* (30, −5) and be *open minded on contentious issues and even conflicting opinions* (38, −4) but keep *the team work* in order *within organizational and hierarchical boundary* (27, −4).

Unlike perspective 1, they underline the importance of *passionate display of senior management leadership from contractor* (8, +4) to establish the right working culture. These practitioners give a different meaning to the project team as it are mainly the contractor's people who actually do the work. They quite strongly believe that *contractor's confidence on the owner's reliability and trustworthiness* (3, +3), *specifying targeted performance in the contract* (35, +3), *reward and recognition to people's performance and behavior* (37, +3), and *passionate display of leadership from the owner's senior management* (55, +3) are important means to support the project team. Reflections from defining respondents encapsulate this core belief:

> "The relationship works on trust and trust grows on transparency just like a marriage (R12 and R24). "Shared objectives is the basis for couple structure, you always explain every problem with your counter-part" (R12 and R16). "Social interaction or celebration is necessary to build trust and you need this to motivate people" (R12, R16, and R30). "Tensions can be helpful and at the same time you need a structure and clarity because you have to work with many people and complex works" (R6, R12, R24, and R30). "Basically it is people who do the project; you need leadership from both sides to provide structure and necessary resources to the project team" (R16 and R17).

Unlike other perspectives, they particularly disagree that *contractor's early involvement* (12, −3) and *owner's effort in FED* (*front-end development*) (13, −3) will have a positive effect on the relationship. They argue that "the early stage of a project requires a different set of attitudes and skills which are different from [execution] contractor"(R16). "It is nice to have contractor involvement early on and owners doing sufficient front-end development, but we can always start from scratch and build it along the way" (R12 and R30). Their strong

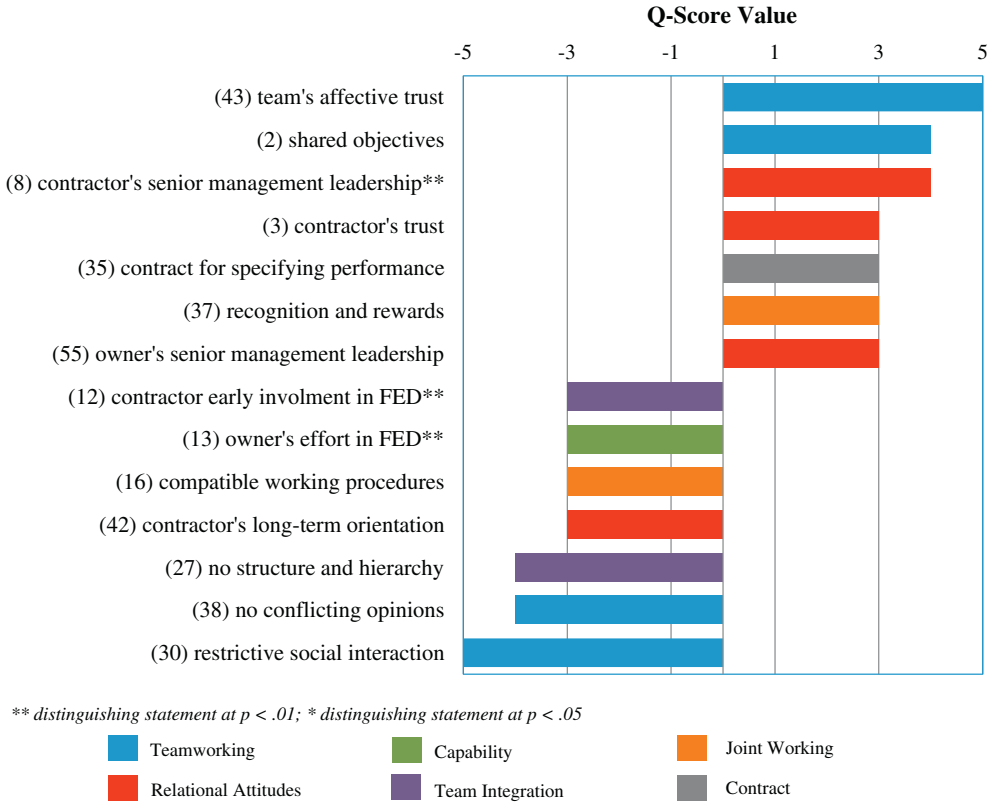


Fig. 3. Perspective 2 (execution focused team): defining and distinguishing statements.

rejection of *contractor's long-term orientation* (42, +3) further illuminates this perspective as execution-oriented view.

4.3.3. Perspective 3 (joint capability and structure)

The third perspective labeled *joint capability and structure* accounts for 12% of the explained variance and is defined by 5 practitioners (2 owner project managers and 3 contractor project managers). Practitioners with this perspective particularly emphasize the joint capabilities and structured approach to improve the working relationship. As shown in Fig. 4, they specifically emphasize *owner's efforts in front-end development* (13, +5) and *owner's sufficient competent personnel* (45, +4). *Structure and hierarchy* (27, −5) should be, in fact, in place to govern the relationship. *Conflicting opinions in the project team* should be nurtured (38, −4) because it is important to understand the underlying reason of the differences. Specifying *roles and responsibilities in the contract* (20, +4) will strongly help to clarify the project governance structure.

The underlying arguments of this core belief are expressed in the following reflections:

> "Good front-end means a success at the start and this is what owner must put sufficient effort in. The owner has the knowledge about the assets and this knowledge must be made available to the project. The owner with the help of contractor can prepare these in advance" (R8 and R9). "To some degree a project team is just like a military team" (R15). Without boundary and hierarchy, the structure is missing and therefore there is no basis for cooperation and decision making process" (R7 and R9). "Structure comes with a framework to clarify high level tasks, responsibilities, and authority" (R8). "[However], there must be room for conflicts; cherish, understand, solve and decide; it offers an opportunity to create a strong bond and colorful team" (R4, R8 and R9).

In line with the above reasons, these practitioners particularly believe in the importance of *contractor's early involvement in the front-end development* (12, +3) and *contractor's project management capability* (18, +3). They also perceive that *risks have to be managed jointly* (40, +3), *trust within the project team* (43, +3) should be nurtured, and *social interaction* should not be restricted (30, −3). Moreover, integrating the owner's and the contractor's key personnel *into one single team* (33, +2) will further enhance the working relationship.

The need to have a proper project team and relationship structure is found by these practitioners to govern the relationship with certainty but does not need *compatible working procedures* (16, −4). "Compatible systems and procedures are absolutely unnecessary because there are many ways to link your tasks" (R7). "In fact, system and procedure must be established based on the project requirements" (R8). With such an approach, *future offers from the owner* (34, −3) and *from the contractor* (41, −3) are irrelevant

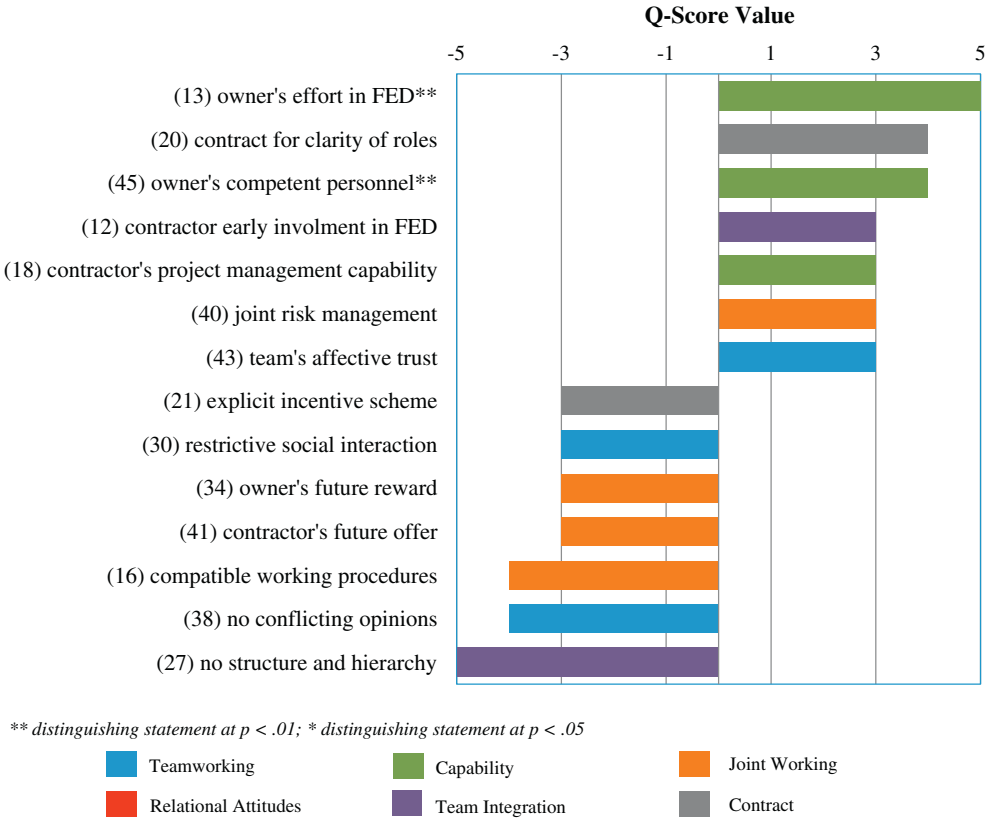


Fig. 4. Perspective 3 (joint capability and structure: defining and distinguishing statements.

since no certainty can be secured. Similar to other perspectives, *explicit incentive scheme* (21, −3) is perceived negatively as "nothing to do with the relationship" (R7) and "it can even be harmful because it distracts people from their work" (R15).

4.3.4. Perspective 4 (senior leadership pair)

The fourth perspective labeled as *senior leadership pair* accounts for 13% of the explained variance and is defined by 4 practitioners (2 owner project managers and 2 contractor project managers). The practitioners with this perspective uniquely underline the visibility of the leadership pair to determine the best formation of the project team and to shape the collaborative working culture. Fig. 5 shows characterizing and distinguishing statements of this perspective.

This core belief emerges from the importance of *consistent display of passionate leadership from owner's* (55, +5) and *contractor's senior management* (8, +4). Similar to other perspectives, they emphasize *trust* (43, +4) and *social interaction* (30, −4) *within the project team*. The following reflections illustrate the core idea of this perspective:

> "Senior management of both parties must focus on the best formation of the project team so the required conditions for the project team are optimal" (R1). This [leadership pair] provides vision and determines the culture of the whole project organization from both sides, what you get is the team members are also being passionate and intrinsically motivated" (R5 and R20). "At the end it is the people who work together and make a project, you need mutual trust" (R5 and R20).

Next, similar to other practitioners, they quite strongly believe in the need to establish the right relational attitudes, mainly with a high level of *awareness on cultural differences* between organization and people (6, +3), *no blaming each other but focus on finding solution when problems occur* (28, +3), and *open and honest* communication (44, +3). They also consider the importance of *contractor's early involvement* (12, +3). These supporting perceptions are reflected as follows:

> "Acknowledging cultural differences means you focus to achieve the project goals. No blaming supports openness and in turn develops trustworthiness among parties and people in the project team" (R21). "You must have an agreement on both sides, if you do not talk openly about a problem, it might already be too late" (R20). "Each team has its strengths and weaknesses, you have to be open about it" (R5 and R21), "it is difficult but you must, to be trustworthy" (R20).

Like other practitioners, the practitioners who share this perspective consider that using a *contract as the basis to manage the project activities* (5, −3) might be counter-productive to the relationship because "you cannot define everything in the

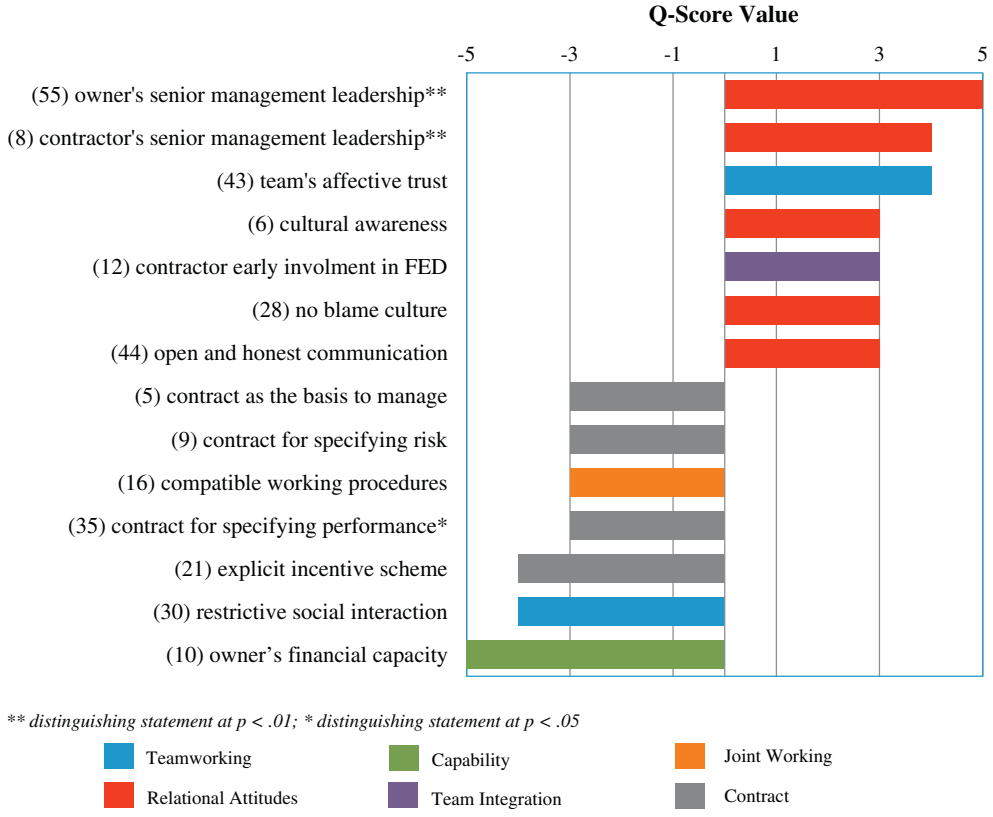


Fig. 5. Perspective 4 (senior leadership pair): defining and distinguishing statements.

contract" (R21). They also disagree that using a contract to specify *explicit incentive scheme* (21, −4), *performance* (33, −3) and *risks* (9, −3) can improve the working relationship. Particularly an explicit incentive scheme is perceived to be "only relevant for senior management" (R5) and "has nothing to do with the people who actually perform the works" (R5 and R21). Furthermore, they strongly believe that *owner's strong financial capacity* (10, −5) has nothing to do with the working relationship. "More money does not mean more passion, besides that the owner can let us know if they cannot pay as much as we ask" (R20 and R21).

4.4. Rank of statements across perspectives

To get an overall rank of statements across four perspectives, we averaged the Z-scores of each statement on four perspectives. Fig. 6 highlights ranks of top (most positive) 15 and bottom (most negative) 15 statements according to the mean Z-scores.

On the top 15 statements, *affective trust* within project team (43, from +1.55), *open and honest communication* (44, from +1.23), *shared objectives* (2, +1.22), and *no-blame culture* (28, +1.11) ranked the top 4 and thus perceived as key ingredients to an effective relationship. Next statements are related to capability such as *contractor's project management capability* (18, +0.78), *competent personnel* of contractor (54, +0.60) and owner (45, +0.57). According to the practitioners, sufficient contractor's capability is "a precondition to a strong project team, mix of people who know what they do".

Next statements are related to senior management such as *owner's* (55, +0.73) and *contractor's* (8, +0.64) *senior management leadership*, *senior management involvement in handling*

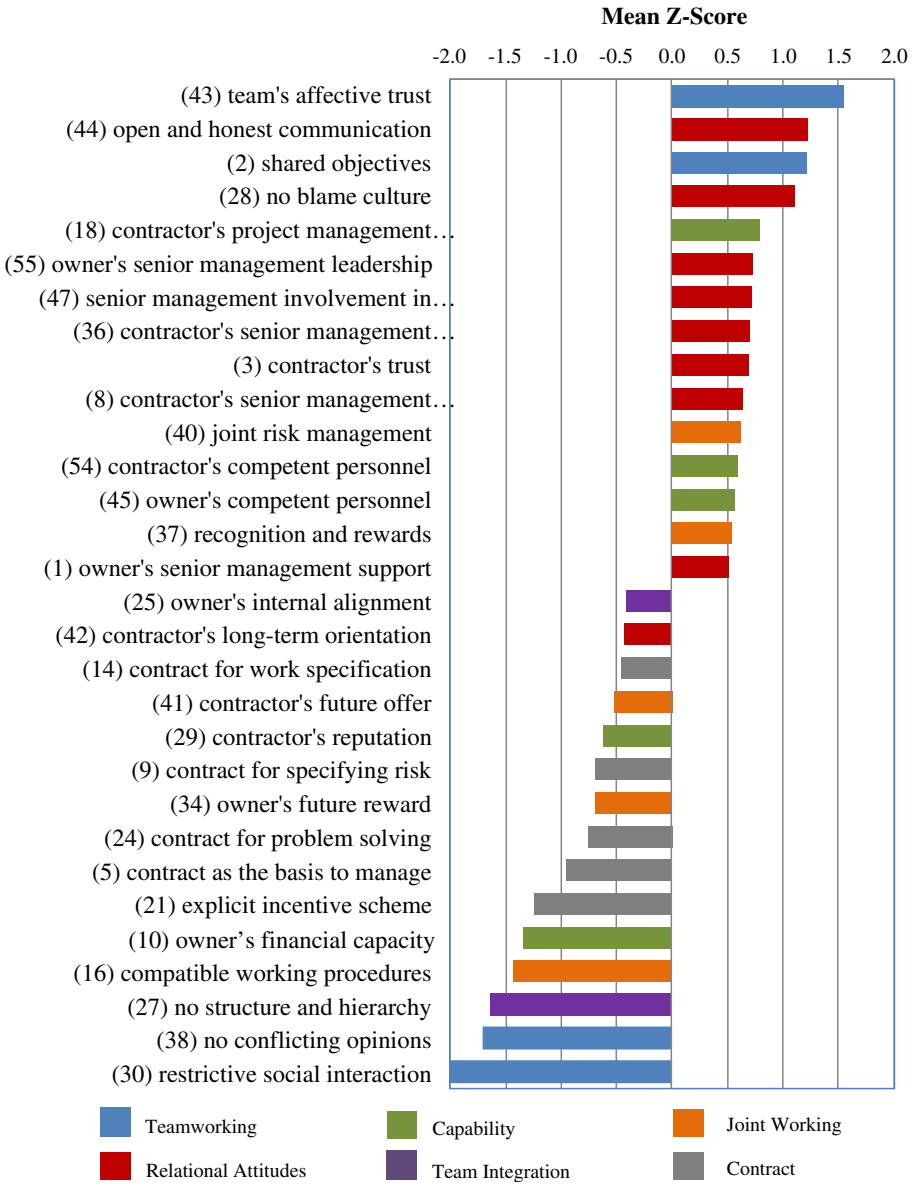


Fig. 6. Top 15 and bottom 15 statements across perspectives.

conflicts (47, +0.72), and *owner's* (1, +0.51) and *contactor's* (36, +0.70) *senior management support*. In their refection, practitioners consider the importance of senior management role in providing "proper and sufficient resources are a good start for good relationship". Finally, the rest statements, *contractor's trust* (3, +0.69), *joint risk management* (40, +0.62), *recognition and rewards* (37, +0.55), and take on the next ranks. *Joint risk management* is acknowledged as "a crucial daily job and by doing it jointly the project team will get the bigger picture" to perform. *Recognition and rewards* are considered important to "make people feel being respected and motivated on their achievement".

On the bottom 15 statements, *restrictive social interaction* (30, −2.21), *no conflicting opinions* (38, −1.71), and *no structure and hierarchy* (27, −1.64) rank at the bottom three. During post-sorting interviews, the practitioners perceived social interactions, conflicts, and structure as elements that persist in the relationship and should not be overlooked. In other words, the reverse of these statements are perceived as important to the relationships.

Next, *compatible working procedures* (16, −1.44) and *owner's financial capacity* (10, −1.34), most contractual functions, i.e.: *explicit incentive scheme* (21, −1.24), *contract as the basis to manage project activities* (5, −0.95), *contract for problem solving* (24, −0.75) *contract for specifying risks* (9, −0.68) and *scope of work* (14, −0.46), *owner's future reward* (34, −0.70), *contractor's future offer* (41, −0.53), *contractor's long-term orientation* (42, −0.43), and *owner's internal alignment* (25, −0.41) are considered to be relatively negative or at least irrelevant to the relationship.

In their reflection, for example, practitioners consider *owner's financial capacity* as precondition to every owner to start a project. While *future offer* and *reward* are understood as long-term relationship but considered irrelevant to current project-based practice to the owner, "Every new project should always start over without any preferences to a particular contractor. Such things are only relevant when we are talking about a program". In one-off projects this means "just talk and a fake relationship". Also regarding the use of contract for specifying problem solving procedure and risks, the practitioners reflect as follows: "In true relationship, once you signed the contract, you put it in the drawer. If things go wrong and you refer to the contract to solve the problems, you will lose the relationship. Contract is a document; risks are dynamic and not one-off definitions that need to be managed beyond the contract. You could never prescribe how to solve a problem and how to mitigate risks in a contract".

5. Discussion

In this study we selected 30 Dutch practitioners on the basis of company (owner or contractor) and role (project director, project manager, and project engineer). Furthermore, three additional pieces of background information were used: educational background, years of experience in projects and their most recent contract type. Eventually, the practitioners' background information did not turn out to be relevant in any of the distinguished perspectives.

5.1. Four perspectives to improve owner–contractor relationship

Each of the perspectives discussed here is based on a particular view within which six conceptual categories were understood by the practitioners. Table 2 summarizes the link between the six conceptual categories and the four perspectives.

The *shared team responsibility perspective* is built on a view that owner–contractor relationship can be improved by focusing on the forming and governing of effective *teamworking*. The core argument of this perspective is placed on teamworking so that key personnel from both sides personally engaged towards a common vision, collectively feel responsible, support and trust each other. Consciously pre-established *relational attitudes* (such as no blame culture, open and honest communication, and mutual respect) at the inter-firm level are perceived as more effective to govern the working relationship. To this perspective, leadership is only meaningful when it is displayed by the project manager who is responsible in day-to-day project activities and not by senior management. Moreover, this perspective considers *joint working* such as performance management, joint risk management, and recognition and rewards program as positive practices to support the teamworking. *Team integration*, in particular, horizontal and vertical structures also can be helpful for check-and-balance mechanisms and clarify the team members' roles and responsibilities. However, too much relying on contract might be harmful. Contract is an absolute necessity in business/commercial sense but not an effective instrument for managing the working relationship. The same attribution also applies to *capability* factor where both owner's and contractor's project management capabilities, technical capabilities, and personnel competencies are perceived as key pre-requisite (given factors) but not sufficient to ensure effective teamworking.

The *execution focused team perspective* is built on a view that owner–contractor relationship can be improved by *teamworking* that is built upon trust and common vision. The effectiveness of the teamworking is, however, primarily driven by the contractor with the focus on project execution. To this perspective, both parties need to establish a proper set of *relational attitudes*. Unlike the previous perspective that rejects the role of senior management; senior management leadership and support are perceived important to secure necessary resources for the project. Moreover, it is also considered that the owner needs to acknowledge contractor's trust and commercial interest. Within *team integration* factor, horizontal and vertical structures are perceived as important feature for check-and-balance mechanisms and clarify the team members' roles and responsibilities. However, early contractor's involvement in the front-end phase is perceived unnecessary and potentially complicates what comes later in execution phase (any changes or extra works). For the sake of the focus on execution, c*ontract* is of importance to teamworking to set the performance target. Regarding *joint working*, only recognition and rewards program is considered useful. Similarly for *capability* factor, only contractor's *capability factors* are considered relevant to the working relationship.

The *joint capability and structure perspective* underlines the formation of the project team at the front-end of the project as the core elements of owner–contractor collaborative relationship.

Table 2
Linking conceptual categories to the practitioners' perspectives.

	Shared team responsibility	Execution focused team	Joint capability and structure	Senior leadership pair
Teamworking	[1] Key personnel from both owner and contractor must share a common vision (+5), respect divergent opinions* (−5), open social interaction* (−4), and personally engage in the project goals (+3), collectively responsible (+3), support (+3) and trust (+2) each other.	[1] The team is the people who actually execute the project. They must trust each other (+5), share a common vision (+4), open social interaction* (−5), respect divergent opinions* (−4) and focus on delivering the project (+2).	[2] The owner should establish the team and involve the contractor as early as possible on the basis of trust (+3), respect divergent opinions* (−5) open social interaction* (−3), and share a common vision (+1).	[2] The team should be formed from owner and contractor key personnel based on trust (+4) and open social interaction* (−4).
Relational attitudes	[2] No blame culture (+4), openness and honesty (+4), and respect cultural differences (+3) are very important. Leadership should be displayed by project manager and team members (*team-shared leadership*) instead of senior management (−3 and −1).	[2] No blame culture (+2) and openness and honesty (+1) are helpful. Senior management pair leadership (+3 and +4) and support (+1 and +2) are important to secure necessary resources for the project.	[4] Senior management pair leadership (+2 and 0) and support (+1 and +2) can be useful. No blame culture (+1) and openness and honesty (+1) are helpful but not always necessary.	[1] Senior management leadership from both sides (+5 and +4) is the key to proper formation of the project team and shape the right working culture. No blame culture (+3), respect cultural differences (+3), and openness and honesty (+3) are important.
Capability	[4] Owner's effort in front-end development (−1), both owner's and contractor's project management capability (−1 and +1) and personnel (+1 and +1) are not matter to the relationship. Owner's financial capacity (−4) is not relevant.	[6] Only contractor's project management capability (+2) and competent personnel (+2) are necessary. Owner's efforts in front-end development (−3), technical capability (−2), personnel (0) do not matter to the relationship.	[1] Owner must put efforts and allocate sufficient competent personnel (+4) in the front-end development (+5). Contractor's project management capability (+3) and competent personnel (+2) are important.	[4] Owner's efforts in front-end development (+2) and contractor's project management capability (+1) are necessary but not the main factor. Owner's financial capacity (−5) is not relevant to the relationship.
Team integration	[3] Structure and hierarchy* (−2) are necessary for check-and-balance mechanism. It does not matter whether contractor is involved early or not (0).	[5] Structure and hierarchy* (−4) on different levels are very important to avoid chaos and to support decision making processes. Contractor does not need to be involved in the front end development (−3).	[3] Structure and hierarchy* (−5) are crucial to establish stability of tasks, responsibilities, and authorities as well as decision making process. Contractor's early involvement is important to integrate the project team (+3).	[3] Structure and hierarchy* (−1) can avoid chaos. Contractor's early involvement is helpful to set the required team culture (+3).
Joint working	[5] It is useful to practice joint performance management (+2), joint risk management (+2), and recognition and reward program (+2).	[4] It is important to have recognition and reward program (+3). Joint performance management (+1) is useful but not always necessary.	[6] It is important to practice joint risk management (+3). Joint performance management (+1) is useful but not always necessary.	[5] Recognition and reward program (+1) and joint risk management (+1) are useful but not always necessary.
Contractual functions	[6] Too much reliance on contract can be harmful (−1 to −3). Contract is required in business/commercial sense but not for managing the working relationship.	[3] Contract is relevant to the working relationship only to set the performance target (+3), specify the remuneration (+1) or define the parties' roles and responsibilities (+1).	[5] Contract is important to structure the clarity of roles and responsibilities among parties (+4). However, too much reliance on explicit incentive scheme might be harmful (−3).	[6] Too much reliance on contract is dangerous (−2 to −4). Contract is required in any project but cannot be used for managing the working relationship.

Note: numbers between brackets indicate the relative ranks of conceptual categories for each perspective based on the mean Z-scores of corresponding statements; numbers between parentheses indicate Q-sort value of corresponding statements;*originally formulated as negative statements.

This also implies the importance of *capability integration*. Both owner's and contractor's capabilities are considered critical to a successful relationship. Such a joint capability developed from the owner's effort in front-end development and owner's sufficient competent personnel as well as contractor's capability. *Team integration* further enhances the owner's initial effort through contractor's early involvement and alignment of subcontractors. Structure and hierarchy are considered crucial to establish stability of tasks, responsibilities, and authorities as well as decision making process. To this perspective, *teamworking* is still perceived as important aspect as the project team should trust each other and share a common vision. A set of *relational attitudes* such as no blame culture, openness and honesty, and senior management leadership and support are useful but not always necessary. C*ontrac*t plays an important role solely to structure the responsibilities of the parties. Only joint risk management is considered as important *joint working* practice.

Finally, the *senior leadership pair perspective* emphasizes *relational attitudes*, particularly senior management leadership pair to the formation of the project team and shaping the right teamworking culture. To this perspective, full alignment at senior management level is perceived as the key factor to set the desired collaborative atmosphere on the project team. Such a collaborative atmosphere should be based on *teamworking*, in particular, mutual trust and other *relational attitudes* such as trust respect of differences, no blame culture, and open and honest communication. Some *team integration* practices (such as contractor's early involvement) and *capability* factor (such as owner's effort in front-end development) are also perceived important to set the required condition for collaboration. Comparable to shared team responsibility perspective, both *joint working* and *contract* are considered to be of relatively least important to the working relationship.

What light do the results of this study shed on the owner–contractor collaborative relationships? All four perspectives affirm the value of teamworking, relational attitudes, team integration, and capability although there is clear disagreement between perspective 3 and other perspectives particularly about capability factor (perspective 3 advocates capability as of primarily importance and other perspectives place it as supporting factor). A closer look into its meaningful elements shows that the sharpest difference seems to be in whether or not quality of a project's front-end development and team formation as the result of the owner's early effort and contractor's early involvement has major positive influence on the working relationship. This difference may be elucidated by Sanderson's (2012) notion of *farsightedness* within which actors should prepare for the future. Perspective 3 seems to be comparable to Sanderson's *Type B farsightedness* that the future is perceived as uncertain but with predictable pattern and therefore a project's plan alongside the project team and its working relationship should be prepared "right from the start" to anticipate turbulences emerging later during execution. Other three perspectives (1, 2, and 4) are analogous to Sanderson's *Type C farsightedness* that projects inherently involve "processes of social construction and characterized by diverse and competing cultures and rationalities [therefore] a shared culture and governance mechanisms [are needed] to encourage collaborative behavior" (p.437). These perspectives move further beyond current forms of governance (pre-designed collaborative contracts or practices) and suggest alternative governing mechanisms as combinations of elements of teamworking and relational attitudes. Perspectives 1 and 2 emphasize more on teamworking (like shared vision, social interaction, trust, personal engagement) while perspective 4 emphasizes more on relational attitudes (particularly senior management leadership and commitment).

5.2. What do we learn?

It is no surprise that the key relationship elements we identified in this study overlap with the 'principles' and key success factors of relational contracting (Rahman and Kumaraswamy, 2008) and construction partnering (Aarseth et al., 2012; Black et al., 2000; Chan et al., 2004; Davis and Walker, 2008). If we compare our study results with two different empirical studies on the success factors of construction partnering of different timelines, a convergent pattern could be shown in Table 3. One study is Black et al.'s (2000) investigation on the success factors of project partnering based on 78 respondents from the UK construction industry. Another study is conducted by Rahman and Kumaraswamy (2008) on the factors for facilitating teamwork relationship based on 80 respondents from Hong Kong construction industry.

Despite the differences in the ranking, all three studies confirm the importance of *mutual trust*, *open and honest communication*, and *shared objectives*. Another convergent result is regarding senior management roles. Whether they are described as 'top management support' or 'senior management commitment' they are all ranked relatively high. In the present study this attention is further enhanced by the focus on 'senior management leadership'. Apparently, this collective view corroborates the findings that the level of leadership from senior management enhances relationship among members of the project team (Baiden and Price, 2011; Chan et al., 2006; Meng, 2011; Yang et al., 2011).

Table 3 also shows a remarkable difference regarding the absence of 'long-term orientation' in the top 10 list of this study compared to the results of Black et al. (labeled as long-term commitment) and Rahman and Kumaraswamy (labeled as long-term perspective). Over time, extant literature has suggested the importance of long-term orientation in inter-firm projects (Boddy et al., 1998; Bosch-Rekveldt et al., 2011; Cheng et al., 2004; Dahlgren and Söderlund, 2001). This study, however, suggests that long-term orientation is not a top priority in existing procurement route that is characterized by one-off contracts (Beach et al., 2005). All statements portraying long-term orientation regarding owner–contractor relationship (Statements 34, 41, and 42) received neutral or negative rank-scores almost throughout. That is, most practitioners perceived long-term orientation as "good metaphor but not realistic… If long-term relationship is desirable then it works only in the context of a program or portfolio of projects". The practitioners perceived that if the relationship is on the basis of one-off projects, long-term orientation is of little concern.

Table 3
Comparing top-10 rank of relationship elements with prior studies.

Rank	This study (2014)	Rahman and Kumaraswamy (2008)[1]	Black et al. (2000)[2]
1	Team's affective trust	Client's top management support	Mutual trust
2	Open and honest communication	Top management support of all parties	Effective communication
3	Shared objectives	Mutual trust among all parties	Commitment from senior management
4	No blame culture	Open communication among all parties	Clear understanding
5	Contractor's project management capability	Enlightened and enthusiastic client	Acting consistent with objectives
6	Owner's senior management leadership	Effective coordination among all parties	Dedicated team
7	Senior management involvement in conflict handling	Teamworking and can do spirit of all parties	Flexibility to change
8	Contractor's senior management support	Long-term commitment to each other for all parties	Commitment to quality
9	Contractor's trust	Clear defined risk allocation/sharing	Commitment to continuous improvement
10	Contractor's senior management leadership	Knowledgeable client about project processes	Long-term perspective

Note: [1]Based on the overall rank of factors facilitating relational contracting (Rahman and Kumaraswamy, 2008, p51); [2]based on the overall rank of factors responsible for successful partnering (Black et al., 2000, p426).

Interestingly, the perception regarding *conflicts within the project team* is considered across perspectives as desirable if *constructive* to the relationship (see Fig. 6). As conflicts are inherent in project organization (Turner, 2009), the practitioners emphasized on managing instead of eliminating conflicts. In their reflections, the practitioners recognized that conflicts should be managed on the basis of *open and honest communication*, *no blame culture*, and *acknowledging cultural differences*. This view reiterates that solving problems and conflicts is pervasive in projects due to competing cultures and rationalities (Sanderson, 2012; Van Marrewijk et al., 2008).

Notably, most contractual aspects seem to be of relatively less influential according to the practitioners. The importance of a contract to *specify roles and responsibilities of the parties* (20, +4) is only highly-ranked in the joint capability and structure perspective. While the contract to *specify targeted performance and its key criteria* (35, +3) is moderately-ranked in the execution focused team perspective. Other contractual functions, i.e.: the contract as *the basis to manage the project activities* (5), *specifying risks* (9), *specifying scope of work* (14), *specifying explicit incentive scheme* (21), *specifying remuneration scheme* (22), and *specifying problem solving procedures* (24) have received mostly negative scores across perspectives. Also what appears from Fig. 6 and Table 2 suggests that working relationship cannot be fully prescribed by contract. This suggests that contract or contractual arrangements were perceived to be relatively less important than other aspects in 'governing' the relationship especially during execution and when external conditions changes. However, this does not eliminate the need for appropriate contractual arrangements. An appropriate contract is necessary, but not sufficient to ensure effective working relationship (Merrow, 2011).

Meng and Gallagher (2012) indicate that contractual incentives have important influence on owner–contractor working relationships. Our findings suggest that explicit financial incentives are less effective in stimulating the intended relationship behavior. In particular, most practitioners warned that too much relying on financial incentives can distract people and the project team from their work. This finding therefore supports prior findings in Sweden (Kadefors, 2004) and Hong Kong (Rahman and Kumaraswamy, 2008) construction partnering that relational attitudes and trust provide a stronger incentive than economic incentive for collaboration and the basis for effective owner and contractor relationship.

Another notable finding is that the statement *all people in the project team work without organizational/hierarchical boundary* (27) is ranked negatively (from −1 to −5) across perspectives. This suggests the opposite preference and contradicts the 'desirable seamless operation without organizational defined boundaries' as important feature of integrated team working (Baiden and Price, 2011; Thomas and Thomas, 2005). All practitioners opposed the concept 'no organizational boundary or hierarchy' as "anarchy and blurring responsibilities". This seems to indicate an inconsistent reasoning. However, a closer view on this finding suggests that both concepts are complementing each other. Seamless operation refers to mutual adjustment and coordination between people in the project team. Meanwhile boundary and hierarchy are required to allocate tasks and responsibilities in the project team.

Furthermore, the findings further clarify how the practitioners perceived joint risk management. Across the four perspectives, *joint risk management* (JRM) was considered from neutral to moderately important (from 0 to +3) to improve the working relationship. It seems that this perception contradicts a shared view that there should be (ex-ante) a clear division of responsibilities (structure) between owner and contractor. According to Rahman and Kumaraswamy (2002, 2004b), JRM is meant as an ex-post *contract* approach to manage unforeseen risks and uncertainties" in addition to foreseen/quantifiable risks ownership stipulated in construction contract. To the respondents, the concept of JRM was understood as "part of the way of thinking because managing a project is mostly managing risks and changes" and therefore "do it jointly to get a bigger picture". We can see the two different but complementary functions of joint risk management and contract's responsibilities of risks.

It seems that the practitioners' perception regarding owner–contractor collaborative relationships are more influenced by their personal values and evolved through day-to-day experiences. On the basis of a construction partnering case study in the Netherlands, Hartmann and Bresnen (2011) suggest that partnering is "an embedded practice and constructed through individuals' social interaction to support unlearning of old routines and behavior". In a similar vein, based on case study of one UK-based construction partnering Chan et al. (2012) indicate

that the actuality of 'doing' partnering involves sense making and informality despite the presence of formal, prescriptive partnering arrangement. The findings of this study indicate that the practitioners do not consider formal partnering or collaborative arrangement as necessary condition as long as its 'principles' can be put into operation. Overall, the view of a collaborative relationship socially constructed and evolving is the winner.

6. Conclusion

Researchers in the field of project management increasingly recognize that pluralism is necessary to study project practices (Söderlund, 2011). This study captured the plurality of project practitioners' view and revealed four preferences for improving owner–contractor relationship, i.e.: shared team responsibility, execution focused team, joint capability and structure, and senior leadership pair. These four perspectives establish the relative significance of teamworking, relational attitudes, team integration, and capability over prescribed joint working procedures and contract. These four perspectives also reflect the views that pluralism is inherent in projects (Söderlund, 2011), projects are about people, their mindsets (Winch and Maytorena, 2011), and characterized by competing cultures and rationalities (Sanderson, 2012).

This study puts a greater focus on the 'actuality of project-based working and management' (Cicmil et al., 2006, p.675) to stimulate a more reflective and developmental approach to understanding project-as-practice (Blomquist et al., 2010; Hällgren and Söderholm, 2011). We answered the call for new orientation of partnering research in construction from the formal and general conceptualization to the informal and contextual constitution of collaborative working relationships (Bresnen and Marshall, 2002; Bresnen, 2007; Chan et al., 2012). The findings of this study contribute to this new orientation, specifically on what the practitioners, in the absence of formal collaborative mechanisms, perceived to be important for the actual 'doing' of the collaboration in projects. In the absence of formal mechanisms, the practitioners sorted out that *teamworking* and *relational attitudes* (reflected by affective trust, common vision and objectives, open and honest communication, no blame culture, constructive conflict, social interaction, and senior management commitment) are the most essential ingredients to owner–contractor collaboration. Contrary to extant literature, contractual aspects were perceived to be relatively less important for 'governing' the working relationship. Also unexpected, long-term orientation was perceived of relative insignificance to collaborative relationships in one-off projects.

It is important to note here that although we rely on a sample of well-informed practitioners who had been involved in many capital projects outside The Netherlands, the list of essential ingredients of collaborative working relationship in Fig. 6 cannot be generalized to other practitioners managing capital projects in other countries. The findings presented herein neither systematically consider the practitioners' cultural background nor their projects settings. However, the research methodology and the Q-set statements developed for this study provided a basis for investigating practitioners' subjectivity in comparison to literature. Moreover, the methodology and findings of this reported study are expected to be of interest to academia and project practitioners worldwide in terms of potential relational view as an alternative to formal prescriptive partnering approach for facilitating collaborative working relationship.

6.1. Managerial implication

It would be misleading to conclude that one perspective is better than the others or is adequate in every situation, but lessons can be learned from the different perspectives. As reflected from the ranks of elements across perspectives, owner–contractor relationships can be improved by nurturing the team's affective trust and shared objectives, by intentionally establishing no blame culture and open and honest communication on inter-firm level, applying clear boundary and hierarchy within the project organization, and encouraging social activities. An effective relationship is, however, contingent upon situation as constructed by the project actors.

There are some notable disagreements on certain elements such as senior management leadership and contractor's early involvement across perspectives. In practice, however, these disagreements should be considered as complementing rather than contradicting given differences in context.

In the view of limited roles of contract in the relationship as perceived by the practitioners, perhaps the key message here is that appropriate contracts are necessary but not sufficient to ensure effective owner–contractor relationships. The reason for this is that contracts are designed to avoid initial ambiguity but are limited in guiding the parties to interpret problems that emerge later. Relational attitudes and teamworking are more effective on providing a platform in which problems can be resolved.

6.2. Limitation and further research

This study has four key limitations which make further research worthwhile. Firstly, this study has mainly focused on indicating that there are indeed different views among project practitioners on ways to improve owner–contractor relationships. The four perspectives revealed cannot be considered exhaustive representation of all practitioners' views. Secondly, this study provided a snapshot of the project practitioners' views on how to improve their working relationship while the efficacy of owner–contractor relationship on the project performance was not investigated. Thirdly, we have seen contractual functions do not play a significant role on owner–contractor relationship but we need to look into more detail into the influence of different contract types in project-specific context. A follow-up research to address these limitations would be to pursue a survey study with a project as an analysis unit. Key elements of owner–contractor relationship identified in this study will be used as starting point to construct the required survey instrument. Finally, some theoretical observation revealed the interdependence between culture and the nature of working relationship. We will extend this in future research, particularly in different project-specific situations involving owners and contractors from different countries.

Appendix 1. Q-set and corresponding factor scores

No.	Statement	Factor 1		Factor 2		Factor 3		Factor 4		Conceptual & category
		Z-SC	Q-SV	Z-SC	Q-SV	Z-SC	Q-SV	Z-SC	Q-SV	
1	Owner's SM (senior management) provides necessary resources & support	0.02	0	0.47	1	0.72	1	0.84	2	RAT
3	The contractor has confidence that owner is reliable and trustworthy	0.68	2	1.09	3	−0.05	0**	1.02	2	RAT
4	The owner believes that the contractor will make its best efforts	0.34	0	−0.27	0	−0.55	−2	0.51	1	RAT
6	Owner and contractor acknowledge and respect cultural differences	1.33	3	−0.09	0*	−0.69	−2*	1.17	3	RAT
8	Contractor's SM displays consistent and passionate leadership	−0.83	−1**	1.21	4**	0.12	0**	2.06	4**	RAT
19	Owner and contractor focus their efforts on delivering current project	−0.81	−1	0.69	1*	−1.16	−2	−0.09	0*	RAT
28	When problems occur, owner and contractor do not blame each other	2.06	4**	1.00	2	0.33	1*	1.03	3	RAT
36	Contractor's SM provides necessary resources and support to the team	0.11	0	1.01	2	0.95	2	0.74	2	RAT
42	The contractor internalizes the owner's long-term goals as their own goals	0.54	1*	−1.03	−3	−1.02	−2	−0.19	0*	RAT
44	Owner & contractor establish open & honest communication … project lifecycle	1.86	4	0.72	1	0.78	1	1.57	3	RAT
47	Owner's & contractor's SM are proactively involved in … escalated conflicts	0.88	2#	0.74	1#	0.25	0#	1.00	2#	RAT
53	The owner recognizes contractor's commercial interest	0.45	1	1.05	2*	0.04	0	0.33	1	RAT
55	Owner's SM displays consistent and passionate leadership	−1.36	−3**	1.11	3	0.94	2	2.23	5**	RAT
2	All people in the team share a common vision and set objectives	2.26	5	1.95	4	0.57	1	0.10	0	TW
11	The team regularly evaluates each other's roles and responsibilities	1.08	3**	−0.16	0	0.14	0	−0.59	−2	TW
15	All people accept joint responsibilities for the team's achievement	1.60	3**	−0.48	−1*	0.12	0	0.69	1	TW
30	As long as people perform well, social activities/events have to be limited	−1.90	−4	−2.57	−5	−1.37	−3	−2.21	−4	TW
32	All people in the team free to share information/knowledge across boundaries	0.48	1	−0.83	−2*	−0.15	0*	0.87	2	TW
38	No contentious issues and conflicting opinions in the team are allowed	−2.17	−5	−2.03	−4	−2.28	−4	−0.34	−1**	TW
39	The project team's primary concern is to execute the project excellently	0.14	0	1.06	2*	0.41	1	0.22	0	TW
43	All people in the project team trust each other	0.87	2	1.98	5	1.19	3	2.15	4	TW
46	The project team embraces divergent views as … to problem solving	0.32	0	−0.46	−1	−0.53	−1	0.70	1	TW
51	All people in the team are personally engaged towards … goals and outcome	1.24	3**	0.38	0	0.24	0	0.10	0	TW
12	The contractor is involved early during FED	−0.05	0**	−1.39	−3**	1.20	3	1.05	3	TIN
17	The tasks are distributed between owner and contractor rather than duplicated	0.12	0	−0.78	−2	−0.37	−1	0.43	1	TIN
23	The contractor aligns subcontractors and suppliers to the project goals	0.56	1	−0.34	−1	0.91	2	−0.27	−1	TIN
25	The owner aligns its internal functions such as business and operation	−1.28	−3**	−0.35	−1	−0.38	−1	0.37	1*	TIN
26	Owner and contractor build on positive experience from previous projects	0.17	0	0.01	0	−1.16	−2**	0.19	0	TIN
27	All people in the team work without organizational/hierarchical boundary	−0.99	−2*	−2.31	−4	−2.84	−5	−0.40	−1*	TIN
31	The team regularly exercises team building/alignment	0.59	1	0.65	1	−0.25	−1	−0.44	−1	TIN
33	A single project team is formed from owner and contractor's key personnel	0.37	1*	−0.72	−2	1.03	2*	−0.74	−2	TIN
10	The owner has strong financial capacity	−1.79	−4	−0.51	−1	−0.72	−2	−2.32	−5	CAP
13	The owner puts sufficient effort and resources on FED	−0.49	−1**	−1.36	−3**	1.96	5**	0.78	2**	CAP
18	The contractor has strong capability in project management	0.54	1#	1.07	2#	1.17	3#	0.35	1#	CAP
29	The contractor has high reputation and credibility in the market	−1.09	−2#	−0.57	−1#	−0.40	−1#	−0.42	−1#	CAP
45	The owner assigns its people sufficiently and with appropriate … and experience	0.55	1*	−0.12	0	1.95	4**	−0.11	0	CAP
48	The contractor has strong technical capability such as …	0.17	0	0.75	1	0.97	2	0.12	0	CAP
49	The owner has strong technical capabilities	0.09	0	−0.98	−2*	−0.02	0	−0.25	0	CAP
50	The owner has strong capability in project management	−0.12	−1#	−0.22	0#	−0.28	−1#	−0.52	−1#	CAP
54	The contractor has highly skilled and experienced people	0.40	1	1.00	2	1.04	2	−0.06	0	CAP
5	The contract is used as the basis for managing all activities	−1.65	−3	−0.88	−2	0.11	0**	−1.37	−3	CON
9	The risks are clearly specified in the contract	−0.97	−2	−0.35	−1	−0.28	−1	−1.13	−3	CON
14	The contract specifies the statement of work as clearly as possible	−0.74	−1	−0.36	−1	0.28	1*	−1.01	−2	CON
20	The contract clearly specifies roles and responsibilities of the parties	−0.84	−2*	0.76	1	1.26	4	−0.26	0*	CON
21	The contract includes explicit incentive schemes	−1.25	−3	−0.15	0**	−1.72	−3	−1.85	−4	CON
22	The contract specifies remuneration scheme fairly and transparently	−0.41	−1	0.59	1	0.39	1	−0.74	−2	CON
24	The contract includes a structured approach to problem solving	−0.85	−2#	−0.59	−2#	−0.54	−1#	−1.03	−2#	CON
35	The contract specifies targeted performance and its key criteria	−0.69	−1	1.09	3	0.85	1	−1.31	−3*	CON
7	Owner and contractor jointly develop key measures and evaluation	0.65	2	0.63	1	0.37	1	−0.54	−1**	JOW
16	Owner and contractor have compatible systems and procedures	−0.64	−1**	−1.57	−3	−1.79	−4	−1.75	−3	JOW
34	The owner rewards a well-performing contractor with a … chance … next project	−0.90	−2	−0.27	0	−1.20	−3	−0.41	−1	JOW
37	The people's performance and behavior are recognized (non)-financially	0.75	2	1.10	3	−0.04	0	0.37	1	JOW
40	Owner and contractor jointly identify and manage the project risks	0.85	2	0.20	0	1.07	3	0.37	1	JOW
41	The contractor offers competitive solutions for a well-performing owner	−0.02	0	−0.58	−1	−1.24	−3*	−0.28	−1	JOW
52	Owner and contractor jointly develop procedures for conflicts/disputes handling	−0.20	−1#	0.00	0#	−0.33	−1#	−0.70	−2#	JOW

Note: **distinguishing statement, significant at $p < .01$; *distinguishing statement, significant at $p < .05$; #consensus statement, does not distinguish between any pair of factors. Statements with a factor score of −5, −4, −3, +3, +4, or +5 (i.e. those ranked in three outer columns on either side of the score sheet are called characterizing for that factor. &RAT = relational attitude; TW = teamworking; TIN = team integration; CAP = capability; CON = contract; JOW = joint working.

Appendix 2. Characteristics of respondents

Resp.	Position	Company	Total years of experience		Highest education	Latest project profile
			Current job	In projects		
1	PM	Owner A	13	20	Engineering & Business	EPC LS; construction of rectifier substation for a light rail line (NL)
2	PM	Contractor 1	10	40	Electrical Engineering	EPC LS; construction of rectifier substation for a light rail line (NL)
3	PM	Owner B	14	26	Electrical Engineering	EPC LS; revamp of rectifier substation for a Metro line (NL)
4	PM	Contractor 1	30	40	Electrical Engineering	EPC LS; revamp of rectifier substation for a Metro line (NL)
5	PM	Owner C	10	29	Mechanical Engineering	EPCm unit rate; two consecutive refinery turnarounds (NL)
6	PM	Contractor 2	11	28	Mechanical Engineering	EPCm unit rate; two consecutive refinery turnarounds (NL)
7	PM	Owner D	12	32	Industrial Engineering	Project alliance; revamp of waste incineration furnaces (NL)
8	PM	Contractor 3	10	30	Mechanical Engineering	Project alliance; revamp of waste incineration furnaces (NL)
9	PM	Owner E	23	31	Engineering	EPC LS; Relocation and construction of a chemical plant (NL)
10	PD	Owner F	12	34	Mechanical Engineering	Project partnering; construction of a new chemical plant (DE)
11	PM	Contractor 4	3	18	Mechanical Engineering	Long-term alliance; petrochemical plant turnarounds (EU)
12	PM	Contractor 4	3	32	Process Engineering	Long-term alliance; modification of natural gas production units (EU)
13	PM	Owner F	7	16	Business & Engineering	EPC LS; construction of a new chemical plant (CN)
14	PE	Owner G	1.5	3	Mechanical Engineering	EPCm CLS; construction of a pilot process intensification plant (NL)
15	PM	Contractor 5	3	12	Mechanical Engineering	PMC - EPC LS; expansion oil storage and pipelines (EU)
16	PM	Owner G	12	16	Mechanical Engineering	EPC LS; unspecified
17	PD	Owner G	6	28	Mechanical Engineering	EPCm unit rate; construction and expansion of chemical complex (S.E.A)
18	PM	Owner G	6	19	Mechanical Engineering	EPC LS; construction and expansion of natural gas plant (RU)
19	PD	Owner H	9	20	Chemical Engineering	Not specified
20	PM	Contractor 6	7	25	Mechanical Engineer	Not specified
21	PM	Contractor 6	1	16	Mechanical Engineering	Not specified
22	PM	Owner G	9	16	Industrial Engineering	EPC LS; construction of offshore oil & gas platform (S.E.A)
23	PD	Contractor 7	13	28	Business & Management	EPC LS; construction of offshore drill ships (NL)
24	PM	Owner G	15	26	Engineering & Business	EPCm CLS; detail engineering design of offshore gas plant (AU)
25	PD	Contractor 8	4	8	Industrial Engineering	EPC LS; construction of offshore wind farms (NL)
26	PM	Contractor 9	15	24	Mechanical Engineering	PMC–EPCm; oil Refinery turnarounds (NL)
27	PD	Owner I	6	27	Mechanical Engineering	Project alliance; petrochemical plant turnaround (NL)
28	PD	Contractor 9	11	23	Civil Engineering	EPC LS; Construction of offshore wind farms (NL)
29	PE	Contractor 8	1.5	5	Offshore engineering	EPC LS; construction of an offshore pipe lay ship (NL)
30	PE	Contractor 10	3	14	Mechanical Engineering	EPC LS; construction of mining facilities (S.A.)

Appendix 3. Factor loadings for four-factor solution

Respondent	Factor 1	Factor 2	Factor 3	Factor 4
1	0.287	−0.039	0.076	0.788*
2	0.612*	0.045	0.028	0.353
3	0.694*	−0.051	−0.138	0.172
4	0.205	0.234	0.578*	0.152
5	−0.034	0.169	0.284	0.683*
6	0.021	0.680*	0.080	0.015
7	0.030	0.033	0.777*	0.017
8	−0.019	0.274	0.567*	0.144
9	0.181	0.294	0.655*	0.141
10	0.654*	0.314	0.357	0.082
11[#]	0.494	−0.169	0.548	0.293
12	0.145	0.724*	0.064	0.268
13	0.619*	0.006	0.254	0.104
14[#]	0.404	0.248	0.288	0.468
15	0.132	0.226	0.555*	0.479
16	0.146	0.495*	0.149	−0.113
17	0.151	0.560*	0.263	0.434
18[#]	0.384	0.402	0.356	0.205
19	0.675*	0.154	0.155	0.095
20	0.142	0.513	0.164	0.578*
21	0.204	0.060	0.030	0.520*
22[#]	*0.518*	*0.487*	*−0.092*	*0.481*
23	0.682*	0.275	0.029	0.071
24	−0.017	0.542*	0.285	0.310
25	0.682*	0.335	0.203	0.091
26	0.567*	0.224	0.500	−0.003
27[#]	*0.527*	*0.067*	*0.373*	*0.419*
28	0.656*	0.061	0.014	0.538
29[#]	*0.231*	*0.304*	*0.325*	*0.450*
30	0.348	0.608*	0.299	0.124
% **EV**	**18**	**12**	**12**	**13**
Defining sorts	**9**	**6**	**5**	**4**

Note:*indicates a defining sort, i.e.: significance at the .05 level ($f > 0.27$) and $f^2 > h^2/2$. R20 significantly defines Factor 4 because of its highest loading:

- $f_{20;4} = 0.578 > 0.27$ (significant at $p < .05$); and
- $f^2_{20;4} = 0.578^2 = \mathbf{0.334} > h^2/2 = (0.142^2 + 0.513^2 + 0.164^2 + 0.578^2)/2 = \mathbf{0.322}$

[#]indicates a confounding sort, i.e.: its loadings are significant on multiple factors, shown as shaded cells. R29 is a confounding sort to Factors 2, 3, and 4 because:

- $f_{29;2} = \mathbf{0.304}$; $f_{29;3} = 0.325$; and $f_{29;4} = 0.45 > 0.27$ (significant at $p < .05$); but
- it highest loading $f^2_{29;4} = 0.45^2 = \mathbf{0.203} < h^2/2 = (0.231^2 + 0.304^2 + 0.325^2 + 0.450^2)/2 = \mathbf{0.227.}$

Conflict of interest

The authors declare that there are no conflicts of interest.

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